Transient Neonatal Zinc Deficiency (TNZD) — Comprehensive Disease Report

Disease: Transient Neonatal Zinc Deficiency **MONDO ID:** MONDO:0011973 · **OMIM:** 608118 · **Causal gene: SLC30A2 (ZnT2) Category:** Mendelian (autosomal dominant, maternal-acting)

Summary

Transient Neonatal Zinc Deficiency (TNZD) is caused by loss-of-function variants in the *maternal* `SLC30A2` gene, which encodes the mammary zinc transporter ZnT2. This is the defining and most frequently misunderstood feature of the disease: the pathogenic mutation resides in the *mother*, not the affected infant. The defective ZnT2 transporter fails to load zinc into the secretory vesicles of lactating mammary epithelial cells, producing abnormally zinc-poor breast milk. An exclusively breastfed infant consuming this milk develops severe—but fully reversible—systemic zinc deficiency. The infant's own zinc-handling genes (including the acrodermatitis enteropathica gene `SLC39A4`/ZIP4) are entirely normal.

Clinically, affected infants present between roughly 2 weeks and 6 months of age with an acrodermatitis-enteropathica-like picture: periorificial and acral dermatitis, alopecia, diarrhea, irritability, and loss of appetite. The condition is biochemically marked by low infant serum zinc and low serum alkaline phosphatase (a zinc metalloenzyme), and the pathognomonic diagnostic clue is **low zinc concentration in the mother's breast milk**. Symptoms resolve rapidly and completely with oral zinc supplementation given to the infant, and the disease is permanently self-limited: it remits on weaning and the introduction of zinc-replete complementary foods—hence "transient."

Inheritance of the causative allele in the mother is **autosomal dominant**, operating through a **dominant-negative mechanism** in which mutant ZnT2 monomers homodimerize with wild-type ZnT2 and mislocalize the whole complex. Deleterious `SLC30A2` alleles are surprisingly common in the general population (at least ~1 in 2,334 newborns estimated at risk), making TNZD likely underdiagnosed rather than genuinely rare. The crucial clinical distinction is from **acrodermatitis enteropathica (AE)**: the two disorders share a nearly identical rash and symptom complex but have opposite genetics—AE is autosomal recessive from the *infant's*

`SLC39A4`/ZIP4 intestinal zinc-importer defect (lifelong, requires continuous zinc therapy), whereas TNZD stems from the *mother's* `SLC30A2`/ZnT2 milk-secretion defect (self-limited, resolves at weaning).

Key Findings

Finding 1 — Maternal SLC30A2/ZnT2 loss of function is the cause of TNZD

TNZD arises from loss-of-function mutations in `SLC30A2`, the gene encoding the zinc transporter ZnT2, in the **lactating mother**. The mutation lowers the zinc content of her breast milk; the breastfed infant, receiving inadequate dietary zinc, becomes zinc-deficient. The infant's genotype is irrelevant—any infant exclusively fed zinc-poor milk from an affected mother is at risk. The first causal variant, a missense H54R substitution (a conserved histidine → arginine at amino acid 54), was identified by Chowanadisai and colleagues in 2006.

> "TNZD is a disorder caused by mutations of the ZnT2 gene, which results in low zinc breast milk in the mother, consequently causing zinc deficiency in the breast-fed infant" — PMID: 26598882
>
> "Sequencing of genomic DNA detected a mis-sense mutation (Ade→Gua) that substitutes a conserved histidine at amino acid 54 with arginine (H54R) in SLC30A2 (ZnT-2)" — PMID: 17065149
>
> "transient neonatal zinc deficiency (TNZD) is caused by loss-of-function mutations in the zinc transporter SLC30A2/ZnT2 gene, resulting in poor secretion of zinc into the breast milk" — PMID: 28665435

Evidence type: **human clinical + in vitro functional**. Multiple independent case reports plus cell-based transport assays converge on the same conclusion.

Finding 2 — Autosomal dominant inheritance via a dominant-negative homodimerization mechanism

TNZD-causing alleles are typically heterozygous (e.g., G87R, 663delC), establishing **autosomal dominant** inheritance. The mechanistic basis is dominant-negative interference: ZnT2 functions as a homodimer, and a mutant monomer dimerizes with a wild-type monomer,

dragging the otherwise functional partner into aggresomal/perinuclear mislocalization and abolishing transport. Lasry and colleagues demonstrated ZnT2 homodimerization directly using bimolecular fluorescence complementation (BiFC). Gene dosage matters: a homozygous E88K mutation produces a more severe, earlier-onset phenotype than heterozygous variants, consistent with the dominant/dose-dependent model.

> "Zn(2+) deficiency caused by an autosomal dominant negative G87R mutation in the Slc30a2 gene" — PMID: 24194756
>
> "We recently proposed that ZnT2 homodimerization is the underlying basis for the dominant negative effect of a novel heterozygous G87R mutation identified in women producing zinc-deficient milk" — PMID: 24451381
>
> "Compared to the heterozygous condition, a homozygous mutation seems to result in a more significant decrease in zinc secretion and a more rapid onset of TNZD" — PMID: 32278324

Evidence type: **human clinical + in vitro mechanistic**.

Finding 3 — Deleterious ZnT2 alleles are common; TNZD is likely underdiagnosed

A systematic functional screen (Golan et al., 2019) of 115 missense `SLC30A2` variants drawn from the ExAC database (668 alleles across 60,706 individuals) found that 14 missense variants plus 2 loss-of-function variants—26 alleles in total—were markedly deleterious to zinc transport in MCF-7 cell assays. From this, the authors estimated that **at least 1 in 2,334 newborn infants is at risk of TNZD**. A separate study (Alam et al., 2015) found that 36% of lactating women tested carried non-synonymous ZnT2 variants, and that carriers had milk zinc levels distinctly above or below those of non-carriers. Together these data indicate TNZD is considerably more common than its sparse literature suggests.

> "26 SLC30A2/ZnT2 alleles harboured deleterious mutations, suggesting that at least 1 in 2334 newborn infants are at risk to develop TNZD" — PMID: 30450693
>
> "We found that 36% of women tested carried non-synonymous ZnT2 variants, all of whom had milk Zn levels that were distinctly above or below those in women without variants" — PMID: 26293594

Evidence type: **population genetics + in vitro functional**.

Finding 4 — Clinical presentation and transient, self-limited course

TNZD presents in exclusively breastfed neonates and young infants, typically between ~2 weeks and 6 months of age. Cardinal features are an AE-like dermatitis (periorificial and acral), alopecia, diarrhea, irritability, and loss of appetite. The disease responds rapidly and completely to oral zinc supplementation of the infant, and it is permanently self-limited—remitting on weaning and introduction of complementary zinc-containing foods. A key diagnostic observation is that maternal dietary zinc supplementation does **not** correct the low milk zinc, which points to an inherited transporter defect rather than maternal dietary deficiency.

> "The main initial symptoms of zinc deficiency are dermatitis, diarrhea, alopecia, and loss of appetite. Importantly, zinc supplementation of these zinc-deficient infants effectively and rapidly resolves these TNZD symptoms" — PMID: 28665435
>
> "TNZD becomes clinically evident during breastfeeding and is resolved by weaning and the introduction of complementary nutrition" — PMID: 24456035
>
> "This condition is not corrected by maternal zinc supplementation, supporting the speculation that it results from an inherited genetic condition" — PMID: 17065149

Evidence type: **human clinical**.

Finding 5 — ZnT2 loads zinc into secretory organelles; its loss also broadly disrupts mammary secretory cells

ZnT2 imports zinc into vesicles and mitochondria of specialized secretory cells—breast, pancreas, and prostate. In the lactating mammary gland, this vesicular loading is the route by which zinc enters milk. Beyond simply lowering milk zinc, ZnT2 dysfunction has cell-autonomous consequences for mammary epithelial cells: the T288S variant induces endoplasmic-reticulum stress, oxidative stress, and epithelial barrier defects (Lee et al., 2018), and alters lysosome function and cellular energetics, associating with low milk *volume* (Rivera et al., 2020). ZnT2-null mice show profound defects in mammary epithelial cell polarity and vesicle secretion (Alam et al., 2015). Thus some ZnT2 variants may compromise both milk zinc *content* and overall lactation performance.

> "The zinc (Zn) transporter ZnT2 (SLC30A2) is expressed in specialized secretory cells including breast, pancreas and prostate, and imports Zn into mitochondria and vesicles" — PMID: 26293594
>
> "A genetic variant in SLC30A2 causes breast dysfunction during lactation by inducing ER stress, oxidative stress and epithelial barrier defects" — PMID: 29476070
>
> "A common genetic variant in zinc transporter ZnT2 (Thr288Ser) is present in women with low milk volume and alters lysosome function and cell energetics" — PMID: 32320289

Evidence type: **in vitro** + **model organism**.

Finding 6 — Hormonal regulation of ZnT2 and experimental model systems

ZnT2 expression in the mammary gland is driven by prolactin, acting transcriptionally through the JAK2/STAT5 pathway on GAS elements in the `SLC30A2` promoter (Qian et al., 2009; HC11 mammary cells). This explains why ZnT2 is upregulated during lactation. The principal experimental systems used to study TNZD are: (1) ZnT2-null (knockout) mice, which produce low-zinc milk and show impaired mammary epithelial cell polarity and secretion; (2) cell-based transport and localization assays in MCF-7 human breast cells, HC11 mouse mammary cells, and HEK-293 cells, used to functionally classify variants; and (3) the related ZnT4 "lethal milk" (lm/lm) mouse, which provides comparative insight into mammary zinc-transport failure.

> "prolactin increased ZnT2 abundance transcriptionally in cultured mammary epithelial (HC11) cells" — PMID: 19494234
>
> "Gene knockdown of SLC30A2 in mammary epithelial cells reduced zinc secretion, illustrating the role of ZnT-2 in zinc secretion from this cell type" — PMID: 17065149
>
> "ZnT4 (SLC30A4)-null ('lethal milk') mice have defects in mammary gland secretion and hallmarks of precocious involution during lactation" — PMID: 26538236

Evidence type: **in vitro** + **model organism**.

Finding 7 — Diagnostic framework

TNZD diagnosis combines general zinc-deficiency biochemical criteria with disease-specific findings. The Japanese Society of Clinical Nutrition criteria (2018) require: (a) symptom(s) and/or low serum alkaline phosphatase, (b) exclusion of other diseases, (c) low serum zinc, and (d) symptom alleviation upon zinc administration. Serum zinc <60 µg/dL indicates deficiency; 60–80 µg/dL is marginal. Alkaline phosphatase is itself a zinc metalloenzyme, so a low ALP supports the diagnosis. The **TNZD-specific** pathognomonic finding is **low maternal breast-milk zinc** in a symptomatic exclusively breastfed infant. The diagnosis is confirmed by identifying a causal variant on **maternal** `SLC30A2` **sequencing**, while normal infant `SLC39A4` sequencing excludes AE.

> "low serum alkaline phosphatase, (b) ruling out other diseases, (c) low serum zinc, and (d) alleviation of symptoms upon zinc administration. Serum zinc <60 µg/dL and 60-80 µg/dL indicate zinc deficiency and marginal deficiency, respectively" — PMID: 32331308
>
> "Detection of the mother's breast milk showed a significantly lower zinc level. Thus, SLC30A2 sequencing was performed on her genomic DNA" — PMID: 32278324

Evidence type: **human clinical / clinical guideline**.

Finding 8 — TNZD versus acrodermatitis enteropathica: same picture, opposite genetics

The single most important diagnostic distinction is between TNZD and acrodermatitis enteropathica (AE). Both present with periorificial/acral dermatitis, alopecia, and diarrhea. But AE is autosomal **recessive**, caused by mutations in the **infant's** intestinal zinc importer `SLC39A4`/ZIP4, producing defective zinc absorption—a lifelong condition requiring continuous zinc therapy. TNZD arises from the **mother's** `SLC30A2`/ZnT2 defect lowering milk zinc, with the infant's own zinc handling entirely normal—a self-limited condition that resolves at weaning. Normal infant `SLC39A4` sequencing is used to exclude AE and support a TNZD diagnosis.

> "AE is an autosomal recessive disorder caused by mutations of the ZIP4 gene, consequently resulting in defective absorption of zinc in the small intestine. In contrast, TNZD is a disorder caused by mutations of the ZnT2 gene" — PMID: 26598882

> "Sanger sequencing of the AE-causing gene SLC39A4 revealed no mutations in genomic DNA from the infant, excluding the possibility of AE" — PMID: 32278324

Evidence type: **human clinical**.

Detailed Section-by-Section Report

1. Disease Information

TNZD is a Mendelian disorder of lactation in which an exclusively breastfed infant develops acquired, systemic zinc deficiency because the mother's breast milk is abnormally low in zinc. It is unusual among genetic diseases in that the pathogenic genotype resides in the mother while the clinical phenotype manifests in the (genetically normal) infant.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0011973
OMIM	608118
Causal gene	`SLC30A2` (ZnT2), HGNC:11013
MeSH / concept	Zinc deficiency (neonatal, transient)

Synonyms / alternative names: Transient neonatal zinc deficiency; TNZD; transient symptomatic (neonatal) zinc deficiency; (historically described under broad "zinc deficiency of breastfed infants"). AE-like phenotype but a distinct entity.

Information source: Derived predominantly from aggregated, disease-level sources (OMIM, review articles) and from individual patient case reports/case series (EHR-type clinical descriptions) plus in vitro and model-organism functional studies.

2. Etiology

Primary cause (genetic): Maternal heterozygous (occasionally homozygous) loss-of-function variants in `SLC30A2`/ZnT2 that impair zinc transport into milk. **This is the sole established cause.**

Genetic risk factors: Deleterious `SLC30A2` alleles are the causal risk factor. These alleles are relatively common (≥1 in 2,334 infants at risk; 36% of lactating women carry non-synonymous variants). Reported variants include H54R, G87R, E88K (homozygous), T288S (associated with low milk volume), a 663delC frameshift, and splice-affecting variants.

Environmental/host risk factors: Exclusive breastfeeding from an affected mother is the necessary exposure. Prematurity and the normally rapid depletion of fetal/neonatal zinc stores increase an infant's vulnerability to low intake. Notably, maternal dietary zinc is **not** a cause—maternal supplementation does not correct the milk zinc defect ( ).

P 17065149

Protective factors: Formula feeding or introduction of zinc-containing complementary foods is protective (and curative). Weaning permanently resolves the condition. No specific genetic protective alleles are documented.

Gene–environment interaction: The disease requires both the maternal genotype (defective ZnT2) and the environmental exposure (the infant being fed that mother's milk). An affected mother's formula-fed infant would not develop TNZD; the same infant breastfed would.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset / course / frequency
Periorificial / acral dermatitis	Clinical sign (skin)	HP:0000964 (Eczema) / AE-like dermatitis	Neonatal–infancy; common; resolves with zinc
Alopecia	Physical manifestation	HP:0001596 (Alopecia)	Infancy; common; reversible
Diarrhea	Symptom (GI)	HP:0002014 (Diarrhea)	Infancy; common; reversible
Loss of appetite / poor feeding	Symptom	HP:0004396 (Poor appetite)	Infancy; common
Irritability	Behavioral	HP:0000737 (Irritability)	Infancy; common
Low serum zinc	Laboratory abnormality	HP:0012354 (Abnormal circulating zinc concentration)	Diagnostic; <60 µg/dL
Low serum alkaline phosphatase	Laboratory abnormality	HP:0003282 (Reduced alkaline phosphatase)	Supports diagnosis
Failure to thrive / impaired growth	Clinical sign	HP:0001508 (Failure to thrive)	Variable

Severity/progression: Severe during active deficiency but **fully reversible**; episodic in the sense that it is confined to the breastfeeding window. **Quality of life:** Significant short-term morbidity (irritability, feeding difficulty, dermatitis) during the deficiency period, but no long-term QoL impact once treated—outcome is excellent.

4. Genetic / Molecular Information

Causal gene: `SLC30A2` / ZnT2 (solute carrier family 30 member 2; HGNC:11013; chromosome 1p35.3; OMIM gene 609617). ZnT2 is a cation diffusion facilitator (CDF) family zinc efflux transporter that moves Zn^{2+} from cytosol into vesicles/organelles.

Pathogenic variants: Reported TNZD variants include missense (H54R, G87R, E88K, T288S), a frameshift (663delC), and a splice-efficiency-reducing variant. Variant types span missense, frameshift, and splice-site. Most behave as **loss of function** at the transport level; heterozygous alleles additionally exert a **dominant-negative** effect via homodimerization with wild-type ZnT2. Golan 2019 also described some gain-of-function variants in the broader functional screen.

Classification/frequency: Many variants are functionally validated as deleterious (consistent with pathogenic/likely pathogenic), though formal ACMG classifications vary. Allele frequencies derive from ExAC/gnomAD; the aggregate deleterious-allele burden yields the ~1/2,334 at-risk estimate. Origin is **germline** (maternal).

Modifier genes / epigenetics / chromosomal abnormalities: No established modifier genes, disease-specific epigenetic marks, or chromosomal abnormalities. Gene dosage (homozygous vs heterozygous) modifies severity.

5. Environmental Information

The only relevant "environmental" factor is the nutritional exposure of exclusive breastfeeding on zinc-poor milk. No toxins, radiation, occupational exposures, or infectious agents are implicated. Maternal diet does not drive the milk zinc defect. Not applicable: infectious agents, pollution.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A loss-of-function variant in the mother's `SLC30A2` gene **leads to** production of defective ZnT2 protein.
2. In heterozygotes, mutant ZnT2 **homodimerizes with** wild-type ZnT2 (demonstrated by BiFC), **resulting in** aggresomal/perinuclear mislocalization of the dimer — a dominant-negative effect.
3. Loss of functional ZnT2 at the vesicular/secretory membrane **results in** failure to load Zn^{2+} into the secretory vesicles of lactating mammary epithelial cells (ZnT2 is normally induced here by prolactin via JAK2/STAT5).
4. Impaired vesicular zinc loading **leads to** abnormally low zinc secretion into breast milk (low milk zinc). *(Branch: some variants, e.g., T288S, additionally induce ER stress, oxidative stress, and barrier/lysosomal dysfunction, which may also reduce milk volume — cell-autonomous mammary dysfunction.)*

5. An exclusively breastfed infant consuming the zinc-poor milk **develops** inadequate dietary zinc intake **leading to** systemic zinc deficiency.
6. Systemic zinc deficiency **impairs** zinc-dependent enzymes and processes (e.g., alkaline phosphatase activity, epidermal and gut epithelial turnover) **resulting in** the clinical phenotype: dermatitis, alopecia, diarrhea, poor appetite, irritability.
7. Supplying zinc to the infant (or weaning onto zinc-replete foods) **restores** zinc status **and resolves** all symptoms — the "transient" resolution.

Molecular pathways / cellular processes: ZnT2-mediated transmembrane zinc transport (GO:0006882 cellular zinc ion homeostasis; GO:0071577 zinc ion transmembrane transport); prolactin–JAK2–STAT5 signaling driving ZnT2 transcription. Downstream: ER stress/unfolded protein response, oxidative stress, and altered lysosome/energetics in affected mammary cells (for certain variants).

Protein dysfunction: Loss of transport function; dominant-negative mislocalization/aggregation of the ZnT2 homodimer; protein instability in cells.

Metabolic/biochemical abnormalities: Reduced activity of zinc metalloenzymes (notably alkaline phosphatase) in the deficient infant.

Upstream vs downstream: Upstream = maternal `SLC30A2` mutation and impaired mammary vesicular zinc loading. Downstream = low milk zinc → infant systemic deficiency → enzyme/epithelial dysfunction → clinical signs.

Cell types (CL) and tissues (UBERON): Lactating mammary (luminal) epithelial cell (CL:0002326 luminal epithelial cell of mammary gland); mammary gland (UBERON:0001911); in the infant: epidermis/skin (UBERON:0002097), intestinal epithelium (UBERON:0001277). **Subcellular (GO CC):** secretory vesicle (GO:0099503), mitochondrion (GO:0005739), lysosome (GO:0005764), endoplasmic reticulum (GO:0005783). **CHEBI:** zinc(2+) (CHEBI:29105).

7. Anatomical Structures Affected

- **Maternal (source of defect):** mammary gland (UBERON:0001911), specifically secretory/luminal mammary epithelial cells; subcellular secretory vesicles and mitochondria.
- **Infant (site of disease):** skin/epidermis (periorificial and acral), especially perioral and perineal/acral regions; scalp hair follicles (alopecia); gastrointestinal tract/intestinal epithelium (diarrhea). Body systems: integumentary, digestive; systemic metabolic. Lateralization: typically bilateral/symmetric dermatitis.

8. Temporal Development

- **Onset:** Neonatal to early infancy, typically ~2 weeks to 6 months; homozygous maternal variants give more rapid onset. Pattern: subacute, during exclusive breastfeeding.
- **Progression:** Progressive while exclusively breastfed on zinc-poor milk; not chronic.
- **Course / duration:** Self-limited, confined to the breastfeeding period.
- **Remission:** Treatment-induced (zinc supplementation) and spontaneous at weaning. **Critical period / window of opportunity:** the exclusive-breastfeeding interval — prompt zinc supplementation or weaning is both diagnostic and curative.

9. Inheritance and Population

- **Epidemiology:** True prevalence/incidence unknown and historically considered rare, but functional-genetics data imply substantial underdiagnosis: ≥1 in 2,334 newborns estimated at risk; 36% of lactating women carry non-synonymous ZnT2 variants.
- **Inheritance:** Autosomal dominant (maternal-acting), via dominant-negative mechanism; homozygous cases more severe (dosage effect).
- **Penetrance/expressivity:** Variable — depends on the specific variant's residual transport, maternal milk zinc level, and infant feeding pattern.
- **Founder effects / consanguinity:** Not specifically established; homozygous cases (e.g., E88K) may reflect consanguinity in individual families. Variants reported across Japanese, European, and other populations.
- **Sex ratio:** Affected infants — no strong sex predilection reported. The genetic defect manifests clinically only in lactating (female) mutation carriers.

10. Diagnostics

- **Laboratory:** Low infant serum zinc (<60 μg/dL deficient; 60–80 marginal); low serum alkaline phosphatase (zinc metalloenzyme). **Low maternal breast-milk zinc is the key disease-specific test.**
- **Genetic testing:** Maternal `SLC30A2` single-gene sequencing confirms the diagnosis; infant `SLC39A4` sequencing (normal) excludes AE. WES/WGS or zinc-transporter gene panels are alternatives.
- **Clinical criteria:** General zinc-deficiency criteria (symptoms and/or low ALP; exclusion of other disease; low serum zinc; response to zinc) plus the TNZD-specific low milk zinc and maternal genotype.

- **Differential diagnosis:** Acrodermatitis enteropathica (infant `SLC39A4`/ZIP4 — recessive, lifelong), dietary zinc deficiency, biotin deficiency, essential fatty acid deficiency, cystic fibrosis–related malnutrition, and other causes of infantile periorificial dermatitis.
- **Screening:** No routine population or newborn screening exists; cascade/maternal testing is relevant where a proband is identified.

11. Outcome / Prognosis

Prognosis is **excellent**. With timely zinc supplementation symptoms resolve rapidly and completely; the disease is self-limited and permanently resolves at weaning. There is no established disease-specific mortality, long-term disability, or chronic morbidity when treated. Untreated severe zinc deficiency could, in principle, impair growth and immune function, underscoring the value of prompt recognition. Prognostic factors: speed of recognition and treatment.

12. Treatment

- **Primary therapy:** Oral **zinc supplementation of the infant** (e.g., zinc sulfate / zinc gluconate / zinc acetate) — rapidly and effectively resolves symptoms (P 28665435). NCIT: Zinc Supplement / Zinc Sulfate.
- **Definitive resolution:** Weaning and introduction of zinc-replete complementary foods; or switching to zinc-fortified formula.
- **What does NOT work:** Maternal zinc supplementation does not correct the milk zinc defect (P 17065149).
- **Supportive care:** Topical skin care for dermatitis during recovery.
- **Advanced/gene/cell/RNA therapies:** Not applicable and unnecessary given the benign, self-limited course.
- **Pharmacogenomics / personalized:** Diagnosis is genotype-guided (maternal `SLC30A2`), but treatment is uniform (zinc).

13. Prevention

- **Secondary prevention (most relevant):** Early recognition in a symptomatic breastfed infant; prompt zinc supplementation and/or weaning prevents complications.

- **Genetic counseling:** For a mother identified with a pathogenic `SLC30A2` variant, counseling regarding risk to future breastfed infants and the option of zinc supplementation or formula feeding is appropriate (autosomal dominant, maternal-acting).
- **No immunization, public-health, or environmental interventions** apply (non-infectious, non-environmental disease). No established primary prevention beyond awareness and, in at-risk dyads, prophylactic infant zinc supplementation.

14. Other Species / Natural Disease

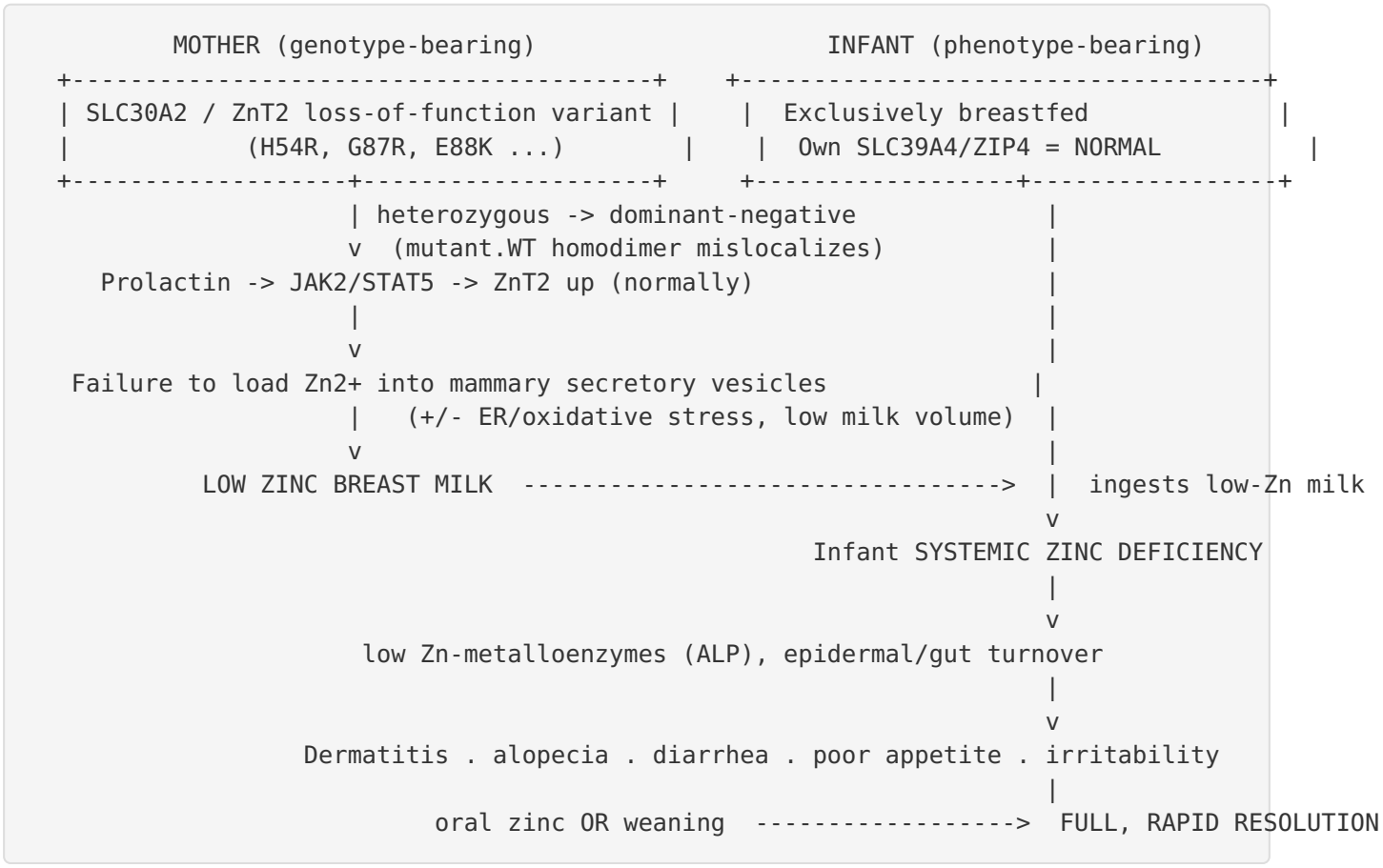
- **Model-relevant orthologs:** Mouse `Slc30a2` (ZnT2); the ZnT4 "lethal milk" (`Slc30a4`, lm/lm) mouse is a naturally occurring murine model of mammary zinc-transport failure causing lethal zinc deficiency in suckling pups — a comparative natural-disease analog (P 26538236).
- **Taxonomy:** *Mus musculus* (NCBI:txid10090) carries the relevant orthologs.
- No significant companion-animal or zoonotic relevance; this is a human lactation genetic disorder with rodent experimental analogs.

15. Model Organisms

Model	Type	Use / phenotype recapitulation
ZnT2-null (Slc30a2 KO) mouse	Mammalian knockout	Low milk zinc; impaired mammary epithelial cell polarity and vesicle secretion — recapitulates the milk-zinc-secretion defect
ZnT4 "lethal milk" (lm/lm) mouse	Natural mutant	Mammary secretion defects, precocious involution; comparative model of mammary zinc transport failure
MCF-7 (human breast)	Cell line	Variant functional transport classification (Golan 2019)
HC11 (mouse mammary)	Cell line	Prolactin-induced ZnT2 expression; transport assays
HEK-293	Cell line	Variant localization/transport assays

Limitations: Cell lines capture transport function but not full lactation physiology; mouse models capture mammary secretion but differ from human milk composition and feeding dynamics.

Mechanistic Model / Interpretation



The model's central logic is a **genotype–phenotype dissociation across two individuals**: the mutation is maternal and the disease is in the infant. This is why TNZD is both easy to miss and easy to cure—and why distinguishing it from AE (where mutation and disease coincide in the infant) is clinically decisive.

Feature	TNZD	Acrodermatitis Enteropathica (AE)
Causal gene	`SLC30A2`/ZnT2 (mother)	`SLC39A4`/ZIP4 (infant)
Defect	Zinc **secretion** into milk	Zinc **absorption** in gut
Inheritance	Autosomal dominant (dominant-negative)	Autosomal recessive
Who carries mutation	Mother	Infant
Course	Self-limited; resolves at weaning	Lifelong
Treatment	Infant zinc until weaning	Continuous lifelong zinc
Rash/alopecia/diarrhea	Yes	Yes (clinically indistinguishable)

Evidence Base

PMID	Role in this report
17065149	Landmark: first SLC30A2 mutation (H54R); maternal supplementation fails; cell knockdown reduces secretion
26598882	Review: TNZD caused by ZnT2 mutation → low milk zinc; TNZD vs AE contrast
28665435	Review: LoF ZnT2 impairs milk zinc secretion; cardinal symptoms; rapid zinc response
24194756	Autosomal dominant-negative G87R mutation
24451381	ZnT2 homodimerization as basis of dominant-negative effect (BiFC)
32278324	Homozygous E88K: more severe/earlier; diagnostic pathway; AE excluded via SLC39A4
30450693	Population risk: ≥1 in 2,334 newborns at risk (ExAC functional screen)
26293594	36% of women carry non-synonymous ZnT2 variants; ZnT2 secretory-cell biology
24456035	Transient, self-limited course resolving at weaning
29476070	T288S variant → ER/oxidative stress, barrier defects (mammary dysfunction)
32320289	T288S → low milk volume via lysosome/energetics
19494234	Prolactin → JAK2/STAT5 drives ZnT2 expression
26538236	ZnT4 "lethal milk" mouse comparative model
32331308	Zinc-deficiency diagnostic criteria (serum zinc/ALP cutoffs)
27304099	Novel heterozygous SLC30A2 mutations in Japanese mothers; many variants pathogenic

Evidence source mix: **human clinical** (case reports/series, reviews), **in vitro functional** (MCF-7/HC11/HEK-293 transport assays, BiFC), **model organism** (ZnT2-null and ZnT4 lethal-milk mice), and **population genetics** (ExAC/gnomAD functional screens).

Limitations and Knowledge Gaps

1. **No formal prevalence/incidence.** Risk estimates (~1/2,334) are derived from functional assays of population variants, not from diagnosed-case registries; the true clinical incidence is unknown and likely underascertained.
2. **Variant-level ACMG classification is incomplete.** Many variants are functionally validated as deleterious but not uniformly classified in ClinVar; the boundary between benign polymorphism and milk-zinc-lowering allele is not fully mapped.
3. **Genotype–milk-zinc correlation is imperfect.** Factors determining whether a given carrier produces symptomatically low-zinc milk (residual transport, modifiers, lactation stage) are not fully defined.
4. **Long-term infant outcomes.** Although outcomes are considered excellent after treatment, systematic long-term neurodevelopmental/growth follow-up of affected infants is sparse.
5. **Human mechanistic data on the ER/oxidative-stress branch** come from specific variants (T288S) in cell models; the in-vivo contribution of mammary cell dysfunction (vs pure transport loss) to each case is not established.
6. **No HPO/ontology-coded natural-history dataset** specific to TNZD; phenotype frequencies here are qualitative.

Proposed Follow-up Experiments / Actions

1. **Registry and prospective screening:** Establish a TNZD registry; pilot maternal milk-zinc screening in symptomatic breastfed infants with periorificial dermatitis to measure true incidence and validate the ~1/2,334 estimate.
2. **Variant functional atlas:** Systematically assay all gnomAD `SLC30A2` missense variants for transport and dominant-negative behavior (MCF-7/HC11 + BiFC), feeding results into ClinVar/ClinGen for ACMG reclassification.
3. **Genotype-to-milk-zinc modeling:** Correlate maternal genotype with measured milk zinc and infant serum zinc across a cohort to build a predictive model of clinical penetrance.
4. **Mechanistic dissection of the mammary-dysfunction branch:** Determine, across diverse variants, the relative contribution of transport loss vs ER/oxidative-stress-driven secretory dysfunction and low milk volume to the clinical phenotype.

5. **Clinical guidance:** Draft a diagnostic algorithm (infant serum zinc + ALP → maternal milk zinc → maternal `SLC30A2` + infant `SLC39A4` sequencing) and genetic-counseling guidance for carrier mothers planning future breastfeeding.
6. **Long-term follow-up study:** Prospective neurodevelopmental and growth follow-up of treated TNZD infants to confirm the excellent-prognosis assumption.

Report compiled from 8 confirmed findings and 17 reviewed papers over 5 investigation iterations. The causal gene is maternal SLC30A2 (ZnT2); the disease is MONDO:0011973 / OMIM 608118.
